INCH-POUND

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SUPERSEDING
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02 May 1991

MILITARY SPECIFICATION

MOUNT, TELESCOPE: 9356166

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for the Mount, Telescope: 9356166 used with the U.S. Army mortar systems (see 6.1). Throughout this specification this instrument is referenced as "mount." This mount has radioactively illuminated scales, level vials, and indices (see 3.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel: General Specification Governimg the Manufacture and Inspection of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government, documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

6523553 - Elevation and Cross Leveling Fixture
7680631 - Dioptometer
7691596 - Azimuth Test Fixture
8658940 - Inspection Aid
9356166 - Mount, Telescope
9356181 - Telescope, Elbow
11750708 - Adapter, Final Inspection, Elevation
11750709 - Adapter, Final Inspection, Azimuth
11750712 - Alignment Gage, Final Inspection, Azimuth

SPECIAL PACKAGING INSTRUCTION

PS 9356166 - Packaging of Mount, Telescope: 9356166

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DLAM 4145.8 - Radioactive Logistics in the DOD Supply System

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)
Part 20: Standards for protection against radiation
Part 30: Rules of general applicability to licensing of by-product material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Nongovernmental standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9356166 all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the manufactured mount shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishing.
c. Part identification and marking.
d. Workmanship.

3.2.1 Radioactive material caution. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement possessing, handling, and shipping of radioactive material.

3.3 Environmental. Any mount subjected to any or all of the requirements of 3.3.1, 3.3.2, and 3.3.3 shall subsequently meet the requirements of 3.4 to 3.8.5 inclusive, unless otherwise specified.

3.3.1 Vibration. The mount shall be vibrated at 30 ± 5 cycles per second frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds. Orientation of the mount shall be in accordance with requirement 3.6. In addition, the azimuth and elevation locking knobs should be turned to a locked position with a fifteen (15) in-lbs torque each.

3.3.2 Shock. The mount shall be shocked six (6) times in each of three (3) positions. The elevation scale shall be set at the 800 mil mark and the azimuth and elevation locking knobs turned to a locked position with a fifteen (15) in-lbs torque each. The applied shocks shall be in a vertical, up direction. In the first position, the mounting dovetail shall be in the horizontal plane and subjected to six (6) shocks, each at 300g for one-half millisecond. In the second and third positions, the mounting dovetail shall be in a vertical plane with its axis at 45° and 225° and subjected to six (6) shocks each at 200g for one millisecond. The shock pulse shall be a one-half sine wave.

3.3.3 Temperature. The mount shall show no evidence of physical failure such as adhesive separations, glass vial breakage, or casting failure after having been exposed and thermally stabilized at -50°F and $+160^{\circ}$F (-46°C and $+71^{\circ}$C).

3.4 External tritium contamination. Tritium in any form shall not be present on the external surfaces of the mount in excess of 1000 dpm/100 sq. cm.

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3.5 Illumination. The markings of the scales, indices, and level vials should be clearly distinguishable when observed with and without a 50% neutral density filter at light levels of zero to ten foot-candles at a distance of 18 to 20 inches from the viewers eye.

3.6 Orientation. The mount shall be positioned as follows to meet the requirements of 3.7 and 3.8. Where line of sight is specified, the requirement shall be met with the line of sight entering the telescope mounting sleeve from the end opposite to the telescope elevation mechanism. A telescope, 9356181, or equivalent dummy load with a suitable adapter, shall be secured by the telescope clamping mechanism.

a. The mounting dovetail surface is in a vertical plane.

b. The axis of the dovetail is vertical.

c. A line of sight provided which is parallel to the above mounting surface and which is perpendicular to the above axis.

d. The elevation and cross level vial bubbles shall be centered within a half of one graduation space.

e. The coarse azimuth scale index indicating 3200 mils within a half of the width of the graduation line.

f. The coarse elevation scale index indicating 800 mils within a half of the width of the graduation line.

g. The indices for the graduated scales on the azimuth and elevation knobs indicating zero within 0.5 mil.

3.7 Performance.

3.7.1 Elevation travel. Rotation of the elevation knob shall cause the line of sight to depress a minimum of 800 mils and elevate a minimum of 100 mils from the position specified in 3.6.

3.7.2 Plumb travel. The line of sight shall track a plumb line within 1.0 mil when elevated and depressed through the range specified in 3.7.1 and the position specified in 3.6. The total lateral spread shall not exceed 1.0 mil.

3.7.3 Lateral displacement. The line of sight shall not exceed 0.5 mil lateral displacement when the elevation motion is reversed by means of the elevation knob.

3.7.4 Vertical displacement. The line of sight shall not exceed 0.5 mil vertical displacement when the azimuth motion is reversed by means of the azimuth knob.

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3.7.5 Horizontal travel. The line of sight hall not vary in elevation in excess of 1.0 mil (total spread of 2 mils) when rotated 6400 mils in azimuth by means of the azimuth knob.

3.7.6 Angular accuracy.

3.7.6.1 Elevation. The angular accuracy with backlash included shall be within 2 mils over the range of elevation travel.

3.7.6.2 Azimuth. The angular accuracy with backlash included shall be within 2 mils over the range of horizontal travel.

3.7.7 Backlash.

3.7.7.1 Elevation. When the amount of backlash at each elevation reading, taken in accordance with 3.7.6.1, is added to the elevation error at that setting, the sum shall not exceed 2.0 mil.

3.7.7.2 Azimuth. When the amount of backlash at each azimuth reading, taken in accordance with 3.7.6.2, is added to the azimuth error at that setting, the sum shall not exceed 2.0 mil.

3.8 Operability.

3.8.1 Telescope clamping mechanism. The telescope clamping mechanism shall be operable in the unlocked and locked positions when using a mandrel test piece which represents the min. metal and max. metal conditions, material, hardness and surface finish of the telescope, elbow: 9356181. These requirements shall be met at the ambient temperature and also when the mount is subjected to the extreme operating temperature requirements of 3.3.3.

3.8.1.1 Unlocked position. The mandrel test piece (max. size) shall rotate freely within its mating diameter and the clamping mechanism shall pivot freely about its axis in a fore and aft direction.

3.8.1.2 Locked position. When locked in place by applying a force of 10 to 12 in-lbs to the wing nut and 15 to 17 in-lbs to the locking lever, there shall be no rotation of the mandrel (min. size) within the inside diameter of the clamp when a 35 in-lb torque is applied. When locked in place with above forces, the clamping mechanism shall not pivot about its axis when a 80 in-lb torque is applied.

3.8.2 Mount locking latch. The mount locking latch shall be operable at the ambient temperature and also when subjected to the extreme operating temperature requirements of 3.3.3.

3.8.3 Locking knobs. After tightening the elevation and azimuth locking knobs to a locked position with a 15 inch-pound torque, there shall be no movement in their respective mechanisms when and 8 inch-pound torque is applied to the elevation and azimuth

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knobs. This requirement shall be met at the ambient temperature and while the mount is subjected to the operating temperatures of 3.3.3. The shift in mount elevation orientation occurring as a result of tightening the elevation locking knob from the unlocked position to the locked position shall be less than 1 mil at ambient temperature.

3.8.4 Azimuth slip scales. The micrometer and coarse azimuth slip scales shall be operable at the ambient temperature and while subjected to the extreme operating temperature requirements of 3.3.3.

3.8.4.1 Torque, coarse azimuth scale. The coarse azimuth scale shall withstand a torque of 3.0 inch-pounds without slippage in both the clockwise and counter-clockwise directions at ambient temperature.

3.8.5 Torque. The running torque in inch-pounds on all knobs shall be within the limits specified in Table I.

TABLE I. Torque values (inch-pounds)

Knob	+60°F to 90°F (+16°C to +32°C) Ambient Temperature		-50°F and +145°F (-46°C and +63°C) Extreme Operating Temperature	
	Min	Max	Min	Max
Elevation	0.5	4.0	0.5	9.0
Azimuth	0.5	4.0	0.5	9.0

3.9 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.10 Workmanship. The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of any defects which affect their function, protective qualities or appearances.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

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4.3 First article inspection.

* 4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer, consisting of three (3) each of the following items, and one each of all remaining items covered by QAPs in TDPL for inspection in accordance with provisions of 4.3.2.

Name	Drawing	QAP	Quantity
Mount, Telescope	9356166		3
Holder Assembly	9356130	9356130	3
Scale Assembly	9356168	9356168	3
Knob Assembly	9356172	9356172	3
Housing	9356174	9356174	3
Bracket Assembly	9356175	9356175	3
Gear	9356190	9356190	3
Knob Assembly	9360178	9360178	3
Shaft Assembly	11741740	11741740	3
Shaft Assembly	11741741	11741741	3

4.3.2 Inspections to be performed. The first article assemblies, components and test specimen shall be subjected to all the examinations and tests specified in Table II, Table III, and their applicable QAPs. It may be inspected for compliance with any or all other requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package List (TDPL). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

a. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with DLAM 4145.8 and Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Tritium contamination	3.4	4.7.4
102. Illumination	3.5	4.7.5
103. Performance	3.7	4.7.6 and 4.7.7
104. Operability	3.8	4.7.8
105. Workmanship	3.10	4.7.9
106. Coarse Azimuth Scale Torque	3.8.4.1	4.7.8.5.1
Minor:	None	N/A

4.5 Special sampling. Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
107. Vibration	3.3.1	4.7.1
108. Shock	3.3.2	4.7.2
109. Temperature	3.3.3	4.7.3
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE IV. Test equipment list.

ITEM	NAME/DESCRIPTION	DRAWING NUMBER
1.	Elevation and Cross Leveling Fixture	6523553
2.	Azimuth Test Fixture	7691596
3.	Adapter, Final Inspection, Elevation	11750708
4.	Adapter, Final Inspection, Azimuth	11750709
5.	Alignment Gage, Final Inspection, Azimuth	11750712
6.	Inspection Aid	8658940
7.	Dioptometer	7680631
8.	Mounting Bracket for Dioptometer	See 4.6.2.3.1
9.	Wall Target	See 4.6.2.3.2

4.6.2.3.1 Mounting bracket for dioptometer. The mounting bracket shall consist of a sleeve and mandrel joined by a connecting arm. The geometric axis of the sleeve and mandrel shall lay in the same vertical plane within .001 inch. The geometric axes of the sleeve and mandrel shall also be horizontal and parallel to each other within .001 inch. The geometric axis of the sleeve shall be 1.50 + .25 inch above the geometric axis of the mandrel. The inside diameter of the sleeve shall be capable of accepting the dioptometer, 7680631 (outside diameter 1.18 - .01 inch). The sleeve shall secure and maintain the dioptometer in a horizontal position. The mandrel shall have a length of 1.000 + .250 inch and an outside diameter of .8435 - .0008 inch. Install the dioptometer in the sleeve of the bracket. Rotate the dioptometer until its vertical reticle line lays in the same vertical plane that contains the geometric axes of the sleeve and mandrel. Lock the dioptometer in this position.

4.6.2.3.2 Wall target. A calibrated wall target shall be utilized in measuring the elevation accuracy requirements of the mount. The wall target shall contain a vertical and horizontal crossline pattern together with limiting displacement marks needed to determine line of sight deviations (azimuth shift) during elevation travels. This wall target shall be calibrated for a measured distance between 6.0 and 9.0 feet, as measured from the mechanical rotational axis (elevation) of the mount.

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4.7 Methods of inspection.

4.7.1 Vibration. Conduct the vibration test with the mount positioned on an adapter affixed to a standard vibrating machine that is capable of providing the amplitude and frequency specified in 3.3.1. The adapter shall position the mount as specified in 3.3.1 during the vibration cycle. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-STD-810. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle, the mount shall be removed from the vibration adapter and subjected to the test in 4.7.4 through 4.7.8. If mount does not comply with any requirements of 3.4 through 3.8, the mount shall be rejected.

4.7.2 Shock. Conduct the shock test with the mount positioned on an adapter affixed to a shock test machine. The adapter shall be capable of holding and positioning the mount as specified in 3.3.2. The number of shocks and applied directions shall be as specified in 3.3.2. The shock pulse shape, peak values, and normal time durations for the positions indicated shall be as specified in 3.3.2. Position the mount (scale and knobs) as specified prior to shock testing. Subsequent to the shock applications (18), the mount shall be removed from the shock test adapter and subjected to the tests in 4.7.4 through 4.7.8. If mount does not comply with any requirements of 3.4 through 3.8, the mount shall be rejected.

4.7.3 Storage/operating temperature. The testing equipment utilized in this test shall conform to the requirements of MIL-STD-810. The mount shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours ± 15 minutes, from room ambient temperature to -50°F. The mount shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure, and while at -50°F (-46°C), the mount shall be subjected to the operability tests in 4.7.8.1 through 4.7.8.6. Upon completion of the low temperature operability tests, the temperature of the chamber shall be raised, over a period of 4 hours ± 15 minutes, to room ambient temperature and the mount allowed to stabilize at this temperature for 2 hours ± 15 minutes, at which time the mount will be subjected to the test of 4.7.8. Upon completion of the room ambient temperature test, the temperature of the chamber shall be raised, over a period of 4 hours ± 15 minutes, to +160°F (+71°C). The mount shall be maintained at this temperature for a minimum of 4 hours. Upon completion of the 4 hour exposure, the temperature of the chamber shall be reduced, over a period of 30 minutes ± 2 minutes, to +145°F (+63°C). The mount shall be maintained at this extreme temperature for a minimum of 2 hours. Upon completion of the 2 hour exposure and while at +145°F (+63°C), the mount shall again be subjected to the operability tests in 4.7.8.1 through 4.7.8.6. Upon completion of the high temperature operability tests, the temperature of the chamber shall be reduced,

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over a period of 4 hours ± 15 minutes, to room ambient temperature. The mount shall then be removed from the test chamber and subjected to the test in 4.7.4 through 4.7.8. If mount does not comply with any requirements of 3.4 through 3.8, the mount shall be rejected.

NOTE: When chamber temperatures are being raised or reduced, the rate of temperature change shall not exceed +2°F (1.1°C) per minute.

4.7.4 External tritium contamination. The contamination test shall be performed by wiping the item under test with filter paper moistened in distilled water. The damp filter paper shall be placed in a container with the proper percentage of scintillation liquid, which is determined by the type of scintillation test equipment used to determine the actual contamination. Insert the sample into this system, count a minimum of 2000 disintegrations and convert the readout into microcurie (222 DPM 10^4 = 1 microcurie). The mount shall be rejected, if the measured contamination is not within the limits specified in 3.4. The liquid scintillating system must be calibrated on six-month intervals, to a standard traceable to an N.B.S. standard.

4.7.5 Illumination. The luminosity test shall be performed in a dark room having a variably controlled light source. To perform this test, the observer must be dark adapted by being in a darkened room for a minimum of 15 minutes before viewing. All scale markings, indices, level vial bubbles, and markings shall be clearly distinguishable when viewed, at a distance of 18 to 20 inches from the viewer's eye, with and without a 50% neutral density filter while the light level is varied gradually from zero to ten foot-candles over a period of one to two minutes. If any of these requirements are not met, the mount shall be rejected.

4.7.6 Orientation, elevation accuracy. The testing equipment specified in Table IV shall be utilized in performing the elevation travel and accuracy tests. The elevation and cross leveling fixture shall be positioned on a vibration free surface and cross leveled in the horizontal plane by means of a certified 10 second level. The Adapter, 11750708, shall be affixed to the elevation and cross leveling fixture and cross leveled by means of the 10 second level so that the interfacing (female) dovetail meets the requirements of 3.6a and 3.6b. Position the mount on the elevation adapter. Orient the mount as specified in 3.6d through 3.6g. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Insert the mandrel portion of the mounting bracket, with dioptometer, into the telescope clamping mechanism of the mount. The eyepiece of the dioptometer shall be on the side of the mount that contains the elevation knob. Rotate the mounting bracket, in the clamping mechanism, until the dioptometer vertical reticle line is plumb. Lock the clamping mechanism. Position the wall target at its calibrated distance, focus the dioptometer on the wall target by means of the objective assembly. Elevate the target until the intersecting point of the horizontal and vertical target line

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coincides to the intersecting point of the dioptometer horizontal and vertical reticle lines. Plumb the vertical target line independent of the dioptometer vertical reticle line.

NOTE: The condition of 3.6d shall be established with backlash removed from the elevation mechanism for rotation in depression.

4.7.6.1 Elevation travel. With backlash removed for direction of travel, and while directing a line of sight through the center of the dioptometer reticle pattern to the vertical target line, depress the line of sight of the mount in 200 mil increments by means of its elevation knob. Return the line of sight through the dioptometer, to the horizontal plane by elevating the elevation and cross leveling fixture. Repeat the above steps until the mount is at maximum depression. Position the inspection aid (8658940) on the gunner's quadrant seats provided on the elevation adapter and verify the mil by measuring the angle established on the elevation adapter. Return the mount and elevation and cross leveling fixture to zero mil elevation. By means of the elevation knob, elevate the line of sight of the mount to its maximum elevation position. Again, by means of the Inspection Aid (8658940) verify the mil value indicated on the elevation scale and micrometer knob of the mount. If the mount does not conform to the requirements of 3.7.1, it shall be rejected.

4.7.6.2 Plumb travel. Perform this test concurrent with the test in 4.7.6.1. Caution shall be exercised to assure that this test is conducted with backlash removed for direction of travel. During the excursions in depression and elevation, if the vertical reticle line of the dioptometer, at the center of the reticle pattern, does not track the vertical target line within the tolerance specified in 3.7.2, as measured on the target, the mount shall be rejected.

4.7.6.3 Lateral displacement. Perform this test in sequence to the test in 4.7.6.1 and 4.7.6.2. With backlash removed for direction of travel and the dioptometer reticle pattern superimposed on the vertical target line as specified in 4.7.7.1, track the vertical target line for approximately 50 mils by means of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob for approximately 50 mils. If lateral displacement of the dioptometer vertical reticle line, as a result of this operation, exceeds the tolerance specified in 3.7.3 as measured on the target, the mount shall be rejected. Tests for lateral displacement shall be performed within 50 mils of each extremity of the elevation scale. A third test point for lateral displacement shall be performed at approximately mid-range of the elevation scale. All points of tests for lateral displacement shall be randomly selected within the above specified areas.

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4.7.6.4 Elevation angular accuracy. Perform this test at 750 mils, 1000 mils, 1200 mils, 1400 mils, and 1550 mils. With backlash removed from fixture rotation in elevation, elevate the adapter to zero mil in elevation. The zero mil elevation position of the elevation adapter shall be verified with the Inspection Aid, 8658940. With backlash removed for rotation of a line of sight in depression, verify that the dioptometer reticle lines are superimposed on the target lines, and that the mount conforms to the orientation requirements of 3.6d through 3.6g. Depress the line of sight of the mount, by means of its elevation knob, to the first randomly selected elevation setting. Then by means of the elevation mechanism of the test fixture, return the line of sight of the mount to the horizontal plane until the dioptometer reticle lines are again superimposed on the target lines without overtravel. Utilizing the inspection aid, measure the elevation angle on the adapter. If the measured angle recorded on the inspection aid does not equal the angle indicated on the scale and micrometer of the mount within the tolerance specified in 3.7.6.1, the mount shall be rejected. While at elevation settings 750, 1200, 1550 lock the elevation locking knob to the specified torque per 3.8.3. Observe the deviation between the dioptometer reticle and target lines. If the deviation or shift exceeds (average of three values) the requirement in 3.8.3, the mount shall be rejected. Observe the location of the level vial bubble. If the position of the bubble is not within the requirement of 3.6d, the mount shall be rejected. Utilizing the foregoing method, proceed to inspect the elevation accuracy of the mount at the three remaining selected elevation settings. Failure at any one of these settings shall be sufficient to reject the mount. The one elevation setting below 800 mils shall be inspected from a zero mil setting of the fixture and a 800 mil setting of the mount. Record the elevation error for each elevation setting tested for use in the elevation backlash test. Caution must be exercised throughout all elevation accuracy measurements to exclude backlash from the mount and test fixture for their appropriate directions of travel.

4.7.6.5 Elevation backlash. Perform this test in sequence to the test in 4.7.6.4. At each of the selected elevation settings, record the elevation (mil) value when the reticle lines are coincident with the target lines. With backlash excluded for direction of travel, override the target line (horizontal) by approximately 50 mils utilizing the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob of the mount. Then, with backlash included, reverse direction of the elevation knob and reestablish coincidence of the reticle and target lines (horizontal), without overtravel. Again record the elevation (mil) value as read on the scale and micrometer knob. The difference between the first and second recorded (mil) value is the amount of backlash. At the selected elevation setting, if the amount of backlash for that elevation position exceeds the tolerance specified in 3.7.7.1, the mount shall be rejected. Perform the elevation backlash test at each of the selected elevation settings tested in 4.7.6.4.

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4.7.7 Orientation, azimuth accuracy. The testing equipment specified in Table IV shall be utilized in performing the azimuth accuracy tests. The azimuth test fixture shall be positioned on a vibration free surface and cross leveled in the horizontal plane by means of a certified 10 second level. Rotate the wheel assembly of the fixture to the zero azimuth setting. Lock the assembly in this position. Verify that the mechanical axis of the target projector collimator is aligned to the pivotal axis of the wheel assembly of the fixture. The height of the collimator shall be such that its optical axis is coincides with the axis of the sleeve of the mounting bracket when the mount, with mounting bracket and dioptometer affixed, is inserted in the Adapter, 11750709. With the Adapter, 11750709, mounted on the wheel assembly, the Alignment Gage, 11750712, together with the dioptometer, shall align the adapter so that its interfacing (female) dovetail meets the requirement of 3.6a and 3.6b. The alignment gage, with dioptometer, shall also be utilized to establish the mounting dovetail surface of the mount parallel to the optical axis of the collimator. Position the mount on the azimuth adapter. Adjust and position the dioptometer and mounting bracket on the mount as specified in 4.7.6. Orient the mount as specified in 3.6d through 3.6g. Verify that the vertical and horizontal reticle lines of the dioptometer and collimator target lines are superimposed. Then ascertain that the optical axis of the dioptometer is horizontal by rotating the wheel assembly of the fixture to 1600 mils, then counter-rotating the mount, by means of the azimuth knob, to return the dioptometer vertical reticle line coincident to the collimator vertical target line. Correct any elevation error, as evidence by the lack of coincidence of the horizontal reticle and collimator target lines, by adjusting the clamping mechanism for elevation of the line of sight. Then rotate the wheel assembly of the fixture to 4800 mils and repeat the above procedure to level the dioptometer. Repeat the entire procedure until the dioptometer is horizontal at 1600 mils, zero mils, and 4800 mils without interim adjustments. Return the fixture to the zero mil azimuth setting and the dioptometer reticle coincident to the target collimator vertical and horizontal lines.

4.7.7.1 Horizontal travel. Verify that the intersecting point of the dioptometer reticle lines coincides with the horizontal target line. With backlash removed for direction of travel, rotate the line of sight of the mount through 6400 mils, in precise 800 mil increments by means of the azimuth knob. At each 800 mil incremental position of the mount, counter-rotate the wheel assembly of the fixture 800 mils to reestablish coincidence of the dioptometer vertical reticle line and vertical target line. At each incremental position, note any lack of coincidence of the horizontal reticle line and horizontal target line. If any vertical displacement of the line of sight through the intersecting point of the dioptometer reticle lines, with reference to the horizontal target line, exceeds the tolerance specified in 3.7.5 as measured on the target collimator, the mount shall be rejected. Position of the bubbles shall be observed and if the requirements of 3.6d are not met, the mount shall be rejected.

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NOTE: Care shall be exercised to assure that backlash is excluded in all horizontal travel measurements. The 800 mil incremental positions of the mount and test fixture shall be accomplished without overtravel.

4.7.7.2 Vertical displacement. This test may be conducted concurrent with the test in 4.7.7.1. At each 800 mil incremental position, reverse the azimuth motion of the mount by means of the azimuth knob a minimum of 50 mils. If as a result of this reversing operation, the line of sight, through the intersection of the dioptometer reticle lines, shifts vertically in excess of the tolerance specified in 3.7.4, as measured on the target collimator, the mount shall be rejected.

4.7.7.3 Azimuth angular accuracy. Perform this test at 800 mil incremental positions (8) as the line of sight of the mount is rotated 6400 mils in azimuth. Verify that the fixture wheel assembly is positioned at zero mil azimuth with backlash removed for appropriate movement in azimuth. With backlash removed for rotation of the line of sight of the mount to 800 mils, verify that the dioptometer vertical reticle line is superimposed on the vertical target line, and that the mount conforms to the requirements of 3.6d through 3.6g. Rotate the line of sight of the mount to 800 mils without overtravel. Counter-rotate the fixture wheel assembly and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Any difference between the mil value recorded on the mount and the scale of the fixture is azimuth accuracy error. Record any azimuth accuracy error for use in the azimuth backlash test. Utilizing the foregoing method, proceed to inspect the azimuth accuracy of the mount at the seven (7) remaining incremental positions. Record the azimuth error for each of the eight (8) incremental positions tested. Observe the position of the elevation and cross level vial bubbles at each position. Caution must be exercised throughout all azimuth accuracy measurements to exclude backlash from the mount and test fixture for their appropriate direction of travel. If the azimuth accuracy of the mount does not conform to the requirements of 3.7.6.2 at any position, the mount shall be rejected. If the position of the bubble does not conform to the requirement of 3.6d at any of the position, the mount shall be rejected.

4.7.7.4 Azimuth backlash. Perform this test in sequence to the test in 4.7.7.3, at each of the eight (8) incremental positions. Subsequent to the azimuth angular accuracy measurements rotate the mount, by means of the azimuth knob, to an azimuth setting of approximately 850 mils. Then, with backlash included, reverse direction of the knob and again obtain coincidence of the dioptometer vertical reticle line and vertical target line without overtravel. Record the azimuth mil value as read on the azimuth knob. Any azimuth change from the original 800 mil setting, as read on the azimuth knob scale, is the amount of backlash. If at the selected azimuth position, the amount of backlash for that position

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added to the azimuth accuracy error recorded for that position, exceeds the tolerance specified in 3.7.7.2, the mount shall be rejected.

4.7.8 Operability. The following test shall be conducted with the mount positioned on a vibration free surface that has a mating dovetail that maintains the mount as specified in 3.6a and 3.6b, and a mandrel that meets the physical requirements specified in 3.8.1. The torque wrench type tester for testing knobs and clamping mechanism torque shall have a range of 0 - 80 in-lbs. The torque wrench type tester for testing rotation of the mandrel shall have a range of 0 - 50 in-lbs.

4.7.8.1 Telescope clamping mechanism, unlocked position. When subjected to a tactile examination, the function of the locking lever shall meet the requirements of 3.8.1.1. When the mandrel is inserted into the telescope holder, it shall rotate freely for 360 degrees and the telescope holder move fore and aft freely. If any of the requirements are not met, the mount shall be rejected.

4.7.8.2 Telescope clamping mechanism, locked position. Insert the mandrel into the telescope holder. Apply a force of 15 to 17 in-lbs to the locking lever and 10 to 12 in-lbs to the wing nut. Apply the specified torques to the mandrel and telescope holder, in turn. If the telescope clamping mechanism does not meet the requirements of 3.8.1.2, the mount shall be rejected.

4.7.8.3 Mount locking latch. Install and remove the mount from the dovetail on the stand a minimum of three (3) times (3 installations). If the mount does not meet the requirements of 3.8.2, at the temperature specified, it shall be rejected.

4.7.8.4 Locking knobs. Rotate the azimuth and elevation locking knobs to a locked position with the appropriate torque wrench and adapter. If the azimuth and elevation mechanisms do not meet the requirements of 3.8.3 when the specified torque is applied to the applicable knob, at the temperature indicated, the mount shall be rejected.

4.7.8.5 Azimuth slip scales. If when subjected to a tactile examination for operability, the micrometer knob slip scales (azimuth and elevation) and azimuth scale does not meet the requirements of 3.8.4, at the temperature indicated, the mount shall be rejected.

4.7.8.6 Torque, coarse azimuth scale. This test shall be performed on Mount, Telescope, 9356166. The sight assembly shall be mounted via the dovetail such that it is possible to level the bubble vials using the azimuth and elevation knobs and with sufficient security to assure that the level will not change during the test. The torque shall be applied in a uniformly increasing manner at a rate not to exceed 1.0 in-lb per second and the actual torque value at which the scale initially slips shall be recorded.

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The test equipment shall not exert a force on the coarse azimuth scale in a downward direction greater than 1.1 lbs while the torque is being measured. Prior to performing the test, the azimuth control knob shall be used to rotate the coarse azimuth scale at least 400 mils in the clockwise direction stopping with the 0 mil mark on the scale aligned with the index arrow. The counter-clockwise torque value should now be obtained. The azimuth control knob should then be used to rotate the coarse azimuth scale 400 mils in the counterclockwise direction and the clockwise torque value should be obtained. The coarse azimuth scale shall then be rotated clockwise approximately 180 degrees with the azimuth control knob, stopping with the 3200 mil mark on the index. Counterclockwise and clockwise torque values shall be obtained at the 3200 mil position in the same manner as that detailed for the 0 mil position. Extreme care must be taken in the design and operation of the test equipment to assure that not even the slightest movement be imparted to the scale in the same direction as the torque is being measured as this could result in erroneous readings. If all four torque values do not equal or exceed the minimum value specified in paragraph 3.8.4.1, the sightunit shall be rejected.

4.7.8.7 Torque, azimuth, and elevation knobs. Utilizing the appropriate torque wrench with adapter, if the running torque required to rotate the knobs throughout their range of rotation is not within the values specified in 3.8.5, for the temperature indicated, the mount shall be rejected.

4.7.9 Workmanship. The mount shall be subjected to a visual and tactile examination for evidence of faulty workmanship. If the mount does not conform to the requirements of 3.10, it shall be rejected.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9356166 and DLAM 4145.8. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mount, Telescope, 9356166, is a component of Sightunit, M67. The mount introduces measured azimuth and elevation readings to Elbow Telescope, 9356181. The sightunit is used with the U.S. Army mortar systems.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).
c. Requirements for submission of first article sample.
d. Applicable stock number.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire control system
Mortar
Sightunit
Tritium

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-AB41)

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